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Aircraft Design – spring 2026

Test 3

1.

- Design flight condition:
 - Freestream:
density 0.3795 kg/m^3 , Mach 0.85, airspeed 252 m/s
 - At inlet front face: Mach 0.63
 - At engine front face: Mach 0.40
 - Mass flow rate through engine: 540 kg/s

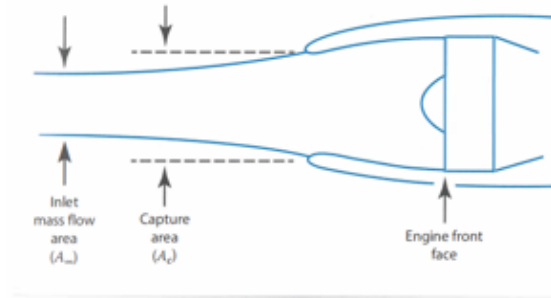


Fig. 10.16 Subsonic inlet capture area.

- Find the inlet mass flow area (A_∞) (in sq m) [10 points]
- Find the inlet diameter (in meters) [15 points]
- Find the engine front face diameter (in meters) [15 points]

$$\dot{m} = \rho V A; \frac{A}{A^*} = \frac{1}{M} \left(\frac{1 + 0.2 M^2}{1.2} \right)^3; \frac{A_{thrt}}{A_{eng}} = \frac{\left(\frac{A}{A^*} \right)_{thrt}}{\left(\frac{A}{A^*} \right)_{eng}}$$

$$A_\infty = \frac{\dot{m}}{\rho_\infty V_\infty} = \frac{540 \frac{\text{kg}}{\text{s}}}{0.3795 \frac{\text{kg}}{\text{m}^3} \cdot 252 \frac{\text{m}}{\text{s}}} = 5.6465 \text{ m}^2$$

$$\frac{A_\infty}{A^*} = \frac{1}{0.85} \left(\frac{1 + 0.2 \cdot 0.85^2}{1.2} \right)^3 = 1.0206$$

$$\frac{A_{inlet}}{A^*} = \frac{1}{0.63} \left(\frac{1 + 0.2 \cdot 0.63^2}{1.2} \right)^3 = 1.1551$$

$$\frac{A_{eng}}{A^*} = \frac{1}{0.4} \left(\frac{1 + 0.2 \cdot 0.4^2}{1.2} \right)^3 = 1.5901$$

$$D_{inlet} = 2 \sqrt{\frac{A_{inlet}}{\pi}} = 2 \sqrt{\frac{1}{\pi} \frac{A_{inlet}}{A^*} \frac{A^*}{A_\infty} A_\infty} = 2 \sqrt{\frac{1}{\pi} \frac{1.1551}{1.0206} 5.6465 \text{ m}^2} = 2.8524 \text{ m}$$

$$D_{eng} = 2 \sqrt{\frac{A_{eng}}{\pi}} = 2 \sqrt{\frac{1}{\pi} \frac{A_{eng}}{A^*} \frac{A^*}{A_\infty} A_\infty} = 2 \sqrt{\frac{1}{\pi} \frac{1.5901}{1.0206} 5.6465 \text{ m}^2} = 3.3467 \text{ m}$$

2.

- Freestream:
density 0.3795 kg/m^3 , Mach 1.40, airspeed 415 m/s
- At engine front face: Mach 0.50
- Mass flow rate through engine: 120 kg/s

Assuming $\gamma = 1.4$

- Assuming a normal shock at the inlet, find the inlet Mach number (just downstream of the normal shock) **[10 points]**

$$M_{inlet}^2 = \frac{1 + \left(\frac{\gamma}{2} - \frac{1}{2}\right) M_1^2}{\gamma M_1^2 - \frac{(\gamma - 1)}{2}} = \frac{1 + (0.2)1.4^2}{1.4 * 1.4^2 - 0.2} = 0.5471$$

- Find the inlet diameter (in meters) **[15 points]**

$$u_{inlet} = u_1 \cdot \frac{2 + (\gamma - 1)M_1^2}{(\gamma + 1)M_1^2} = 415 \frac{2 + (1.4 - 1)1.4^2}{(1.4 + 1)1.4^2} = 245.61 \frac{m}{s}$$

$$\rho_{inlet} = \rho_1 \frac{(\gamma + 1)M_1^2}{2 + (\gamma - 1)M_1^2} = 0.3795 \frac{2.4 \cdot 1.4^2}{2 + 0.4 \cdot 1.4^2} = 1.6897 \text{ kg/m}^3$$

$$D_{inlet} = 2 \sqrt{\frac{A_{inlet}}{\pi}}; \dot{m} = \rho_{inlet} V_{inlet} A_{inlet}$$

$$D_{inlet} = 2 \sqrt{\frac{1}{\pi} \frac{\dot{m}}{\rho_{inlet} V_{inlet}}} = 2 \sqrt{\frac{120}{\pi * 1.6897 * 245.61}} = 0.6067 \text{ m}$$

- Find the engine front face diameter (in meters) **[15 points]**

$$\frac{A_{inlet}}{A^*} = \frac{1}{0.5471} \left(\frac{1 + 0.2 \cdot 0.5471^2}{1.2} \right)^3 = 1.2593$$

$$\frac{A_{eng}}{A^*} = \frac{1}{0.4} \left(\frac{1 + 0.2 \cdot 0.4^2}{1.2} \right)^3 = 1.5901$$

$$D_{eng} = 2 \sqrt{\frac{1}{\pi} \frac{A_{eng}}{A^*} \frac{A^*}{A_{inlet}} A_{inlet}} = 2 \sqrt{\frac{1}{\pi} \frac{1.5901}{1.2593} \frac{120}{1.6897 * 245.61}} = 0.6818 \text{ m}$$

3. Solve this using the simplified magic disk thrust analysis model

$$F = ma = \dot{m}\Delta V = (\rho VS)(V - V_0) = \rho SV(V - V_0) \quad (13.1)$$

$$P_t = FV_0 = \rho SV(V - V_0)V_0 \quad (13.2)$$

$$\begin{aligned} P_{t_{\text{expended}}} &= \frac{\partial \Delta E}{\partial t} = \frac{1}{2} \dot{m} V^2 - \frac{1}{2} \dot{m} V_0^2 = \frac{1}{2} \rho VS(V^2 - V_0^2) \\ &= \frac{\rho S}{2} V(V^2 - V_0^2) \end{aligned} \quad (13.3)$$

$$\eta_{PE} = \frac{P_t}{P_{t_{\text{expended}}}} = \frac{2}{V/V_0 + 1} \quad (13.4)$$

▪ Design flight condition:

- True airspeed: $V_0 = 212 \text{ m/s}$
- Air density: 0.46 kg/m^3
- Thrust requirement: $20,000 \text{ N}$

$$\frac{F}{\rho S} = V^2 - VV_0; 0 = V^2 - VV_0 - \frac{F}{\rho S}$$

$$V = \frac{V_0 + \sqrt{V_0^2 - 4(1)\left(-\frac{F}{\rho S}\right)}}{2}$$

$$\dot{m} = \rho VS$$

$$\eta_{PE} = \frac{2}{\frac{V}{V_0} + 1}$$

Case	Magic disk area [m ²]	Mass flow rate [kg/s]	Accelerated flow speed, V [m/s]	Propulsion efficiency (%)
1	0.4	83.225	452.31	0.6382
2	0.8	133.25	362.09	0.7385
3	1.2	178.77	323.87	0.7912
4	1.6	222.26	301.98	0.8249